NMIMS · MBA TECH · FINANCIAL TECHNOLOGY

Past-Paper Question Bank

Six past papers (2018-19 to 2024-25) analysed and mapped to Units 1–7. Model answers come with a framework, simple language and a memory hook, plus the similar questions likely to be asked.

PART A
What the papers ask: coverage check

PART B
Unit 1–7 questions with answers

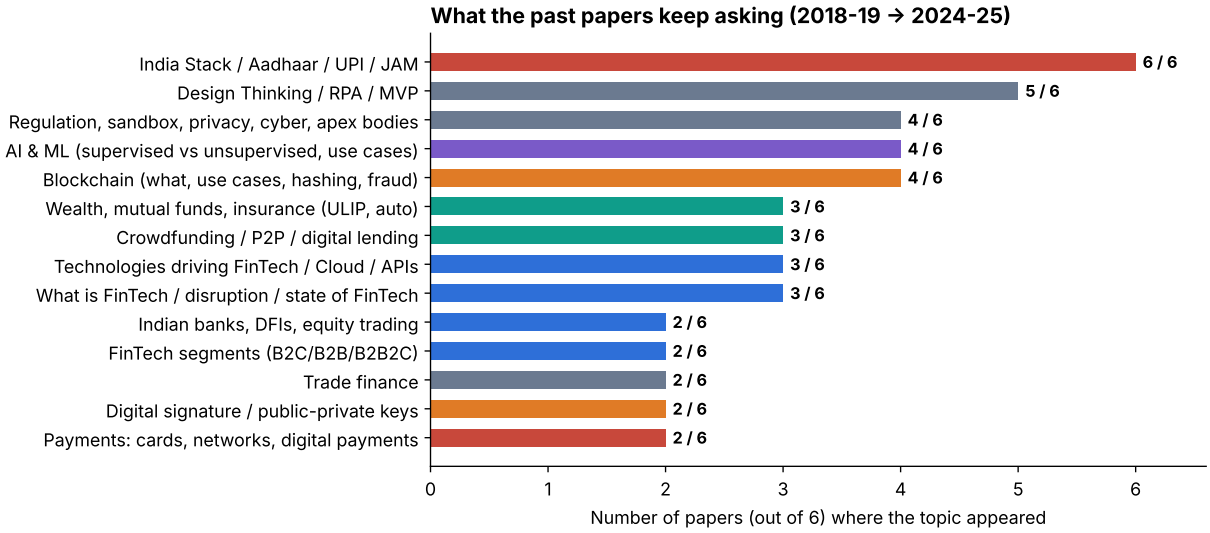
PART C
India-specific & later-unit questions

PART D
Last-minute answer kit

Papers analysed: Final 2018-19 · Final 2019-20 · Final 2022-23 · Final 2023-24 · Re-exam 2023-24 · Final 2024-25. Use with your *FinTech Midterm Study Guide* (Units 1–7).

Part A · What the past papers keep asking

Big finding: the papers are far more **India-focused** than the lecture slides. **India Stack** (Aadhaar, eKYC, UPI, consent) appeared in **all 6 papers**. Blockchain, AI, "technologies driving FinTech" and Design Thinking / RPA also return almost every year. Learn Part C even though it's outside Units 1–7.



EXAM PATTERN (EVERY YEAR)

2 hours · 50–70 marks · **Q1 is compulsory** (often 4–5 short parts of 2–2.5 marks), then **any 4 of the remaining 6** (10–14 marks each). So you need about **5 strong long answers** plus quick definitions. The 2023-24 papers tag each question with course outcomes and Bloom's levels: mostly *understand*, *analyse* and *application*.

READ THIS FIRST · SYLLABUS HAS CHANGED

These papers were set on the **older syllabus**, which was India-heavy: India Stack, RBI and NPCI, Design Thinking, RPA, trade finance. The 2026-27 course follows Michael King's *Fintech Explained* (Units 1–15), so your midterm will lean on Units 1–7 as taught. But teachers reuse favourite questions, and India Stack, UPI and the card payment flow sit naturally inside Units 1, 5 and 7. **Prepare Part B fully, then the top items in Part C.**

Coverage check: have Units 1–7 covered these questions?

Topic asked	Unit	In your study guide?	Where to answer from
What is FinTech, examples, disruption of banking, how it revolutionised finance	1, 2	✅ Fully	Definition, Lens, paradigms, Christensen, 5 dimensions → Q1.1–1.2
FinTech segments; B2C / B2B / B2B2C	1	✅ Fully	Operating segments + 8 lines of business → Q1.3
Technologies driving FinTech; cloud; APIs vs portals	1, 7	✅ Mostly	Foundational technologies table → Q1.4–1.5 (adds RPA, IoT)

Innovation ecosystem (four pillars); state of FinTech in the world vs India	1	🟡 Partly	Ecosystem + investment waves; India numbers added here → Q1.6–1.7
Types of banking institutions in India; DFIs; equity trading process	2	🟡 Partly	Intermediaries list (global); India specifics added → Q2.1–2.3
AI definition, supervised vs unsupervised, AI in finance, daily-life AI	3	✅ Fully	S-U-R, pipeline, ethics, cases → Q3.1–3.2
FinTech in customer onboarding (eKYC) and auto-insurance	3 (+7)	🟡 Partly	InsurTech, telematics, Lemonade, ZhongAn; eKYC added → Q3.3
Robotic Process Automation (RPA)	3 (related)	❌ Not in slides	Added → Q3.4
Blockchain: what, how it works, use cases, fraud reduction, hashing	4	✅ Fully	Hash chain, PoW, Ripple, FTX → Q4.1–4.3
Digital signature; public and private keys	4	🟡 Partly	Keys covered; digital signature added → Q4.4
Trade finance and how FinTech disrupts it	4 / 5	❌ Not in slides	Added → Q4.5
Credit card payment system; acquirer, issuer, network; networks in India	5	✅ Mostly	4-party flow, $100 waterfall; RuPay and NPCI added → Q5.1–5.2
Why digital payments are popular in Indian start-ups; UPI	5	🟡 Partly	Paytm, take-rates, A2A; UPI added → Q5.3–5.4
P2P lending, platform types, credit systems; crowdfunding in India; why start-ups chase digital lending	6	✅ Mostly	Agent vs principal, D-R-E-D, LendingClub; India rules added → Q6.1–6.3
Wealth management transformation; FinTech in mutual funds	7	✅ Fully	Robo workflow, Wealthsimple, Vanguard, hybrid → Q7.1–7.2
Open banking, APIs, Account Aggregator	7	✅ Fully	PSD2, consent flow, BaaS → Q7.3
India Stack (layers, components, benefits, cashless layer), Aadhaar eKYC, JAM	India / 12	❌ Not in Units 1–7	**Part C · C1–C3 (highest priority)**
Govt and RBI initiatives, regulatory sandbox, regulation and enforcement, apex bodies	8, 15	❌ Later units	Part C · C4–C7
Privacy, cyber threats, data security	8	❌ Later unit	Part C · C8

Design Thinking (5 stages), MVP	14	❌ Later unit	Part C · C9
ULIP and regulator; monetary policy; acronym expansions	9 / econ	❌ Not in slides	Part C · C10–C11

Verdict: Units 1–7 fully or mostly cover about **60%** of past-paper marks. The biggest gap is **India-specific digital public infrastructure** (India Stack, UPI, Aadhaar, RBI and NPCI). Part C fills it.

Every question, mapped

Paper	Questions (unit in brackets; ★ = repeated in several papers)
Final 2018-19 60 marks	Digital signature + 3 differences from paper signature (4) · private and public key (4) · expand NBFC, P2P, MoF, RBI, API, GSTN, SEBI, UPI, IRDA, IoT, DLT, ML, NPCI (C) · AI + 2 daily-life examples (3)★ · blockchain + use case (4)★ · 2 apex bodies in India + year (C) · 4 layers of Aadhaar / India Stack (C)★ · JAM (C) · monetary policy objectives (C) · 5 stages of Design Thinking (C)★
Final 2019-20 70 marks	What is FinTech + company examples (1)★ · how it disrupts banking (1, 2)★ · regulatory problems around FinTech (C) · regulatory sandbox in India (C) · India Stack: benefits and components (C)★ · technologies driving FinTech (1)★ · RPA (3)★ · trade finance and FinTech disruption (4)★ · crowdfunding + its future in India (6) · AI + supervised vs unsupervised (3)★ · how AI changes finance (3)★ · blockchain explained (4)★ · blockchain for fraud reduction (4)
Final 2022-23 50 marks	State of FinTech in the world vs India (1) · FinTech segments (1)★ · India Stack benefits (C)★ · short notes: Aadhaar eKYC, UPI (C, 5)★ · technologies driving FinTech (1)★ · short notes: Design Thinking, cloud computing and AI (C, 1)★ · trade finance (4)★
Final 2023-24 50 marks	Why digital payments are popular in Indian start-ups (5) · types of banking institutions in India (2) · four pillars of the innovation ecosystem (1) · hashing (4) · India Stack components with use cases (C)★ · Design Thinking + RPA (C, 3)★ · working of the credit card payment system (5)★ · equity trading: retail and institutional, local participants, FinTech role (2) · FinTech in customer onboarding and auto-insurance (3, 7) · blockchain and its use in FinTech (4)★
Re-exam 2023-24 50 marks	Why start-ups focus on digital lending (6) · ULIP + regulator (C) · B2C vs B2B vs B2B2C (1)★ · digital signature (4)★ · "cashless" layer of India Stack (C, 5)★ · cloud computing; APIs vs portals (1, 7) · acquiring bank, issuing bank, network; networks in India (5)★ · data security and privacy requirements (C) · FinTech in mutual funds and wealth management (7)★ · AI use cases in finance (3)★
Final 2024-25 50 marks	How FinTech revolutionised finance (1)★ · Govt initiatives to promote FinTech (C) · what is regulation and how rules are enforced (C) · Minimum Viable Product (C) · role of DFIs (2) · RBI-pioneered initiatives (C) · India Stack: role and benefits (C)★ · two key blockchain use cases (4)★ · privacy and types of cyber threats (C) · P2P lending, platform types, credit systems (6) · wealth-management transformation (7)★

HOW TO USE PART B & C

Each answer has: ① **a one-line definition** to open with, ② **a named framework** (draw it as boxes), ③ **4–6 points** in plain words, ④ **an example with a number**, ⑤ **a closing line**, and a 🧠 memory hook. Under each answer, "*Also asked as*" lists the variations it covers.

B1 Unit 1 · Introduction to FinTech

Asked in 2019-20, 2022-23, 2023-24, 2023-24 re-exam, 2024-25

10–14 marks **What is FinTech? Explain with examples. How is it disrupting / revolutionising the banking sector?** ★ 3 papers

Definition (opening line): FinTech is the use of software, mobile, cloud, data and AI to deliver financial services **cheaper, easier, faster and more conveniently** than traditional banks. It shifts finance from **product-centric branches** to **customer-centric apps**.

Framework 1 · Examples by line of business

Line	Global example	Indian example
Payments	PayPal, Stripe, Wise	Paytm, PhonePe, Google Pay (on UPI), Razorpay
Lending	LendingClub, SoFi	Lendingkart, KreditBee, BharatPe loans
Wealth	Wealthfront, Wealthsimple, Robinhood	Zerodha, Groww, Kuvera
Insurance	Lemonade, ZhongAn	PolicyBazaar, Acko, Digit
Digital banking	Nubank, Revolut, Chime	Jupiter, Fi (neobanks on partner banks)
Crypto / blockchain	Bitcoin, Ripple	CoinDCX, WazirX

Framework 2 · How it disrupts banks: use the 5 dimensions (I-T-L-R-T)

1. **Information:** alternative data and AI score customers banks can't (thin-file, MSMEs).
2. **Transaction costs:** apps, APIs and UPI cut the cost of a transaction to near zero; no branches needed.
3. **Liquidity:** P2P lending and fractional investing move money without a bank balance sheet.
4. **Risk sharing:** marketplaces spread risk across many investors.
5. **Trust:** moves from bank brands to code, ratings and transparency (e.g. Wise shows its fees upfront).

Christensen angle: FinTechs start at the low end (small payments, small loans, first-time investors), then move upmarket. Banks respond by **partnering** (co-lending, BaaS), **copying** (YONO, iMobile) or **acquiring**. Result: **unbundling → rebundling → collaboration**.

Close with: FinTech isn't killing banks. It forces them to become platforms, while FinTechs depend on banks for licences and deposits.

🧠 **"Cheaper · Easier · Faster · Convenient" + I-T-L-R-T**

Also asked as: "How has FinTech revolutionised the financial industry?" (2 marks: give the definition + the 4 value words + one example) · "Evolution vs revolution in FinTech" · "Is FinTech a threat or an opportunity for banks?"

10 marks **What are the segments in FinTech? Explain each. / Difference between B2C, B2B and B2B2C start-ups.** ★ 2 papers

FinTech can be split two ways: **by who the customer is** (segment) and **by what is sold** (line of business).

Segment	Who pays / who uses	Example	Key challenge
B2C	Sells straight to consumers	Groww, Paytm app, Revolut	High customer-acquisition cost ($500–1,000 per user)
B2B	Sells software / infrastructure to banks and companies, often white-label	Plaid, Razorpay, Perfios, Finastra	Long sales cycles, bank integration
B2B2C	Partners with a bank or brand to reach consumers; keeps visibility of the customer	Credit Karma, PolicyBazaar, BaaS on a ride-hailing app	Shared margins, dependence on partners
C2C / P2P	Individuals deal with each other through a platform	P2P lending (Faircent), crypto exchanges	Trust, fraud, regulation

Lines of business (8): payments · lending · crowdfunding · digital banking · wealth / robo-advisory · InsurTech · crypto · DeFi. Mention RegTech too if the paper says "segments".

🧠 **"Who pays?" decides the segment: consumer (B2C), business (B2B), business for its consumers (B2B2C), each other (C2C)**

Also asked as: "Classify FinTech companies with examples" · "Which segment is most attractive in India and why?" (answer: payments and lending, because UPI and India Stack cut costs)

10 marks **What are the major technologies driving FinTech? Explain each.** ★ 3 papers

Remember **"ABCD + MIRA"**:

Tech	What it is (simple)	FinTech use
AI / ML	Software that learns patterns from data	Credit scoring, fraud detection, chatbots, robo-advice
Blockchain / DLT	Shared, tamper-proof ledger across many computers	Crypto, cross-border payments, trade finance, KYC sharing
Cloud	Renting computing on demand (AWS 2006): SaaS, PaaS, IaaS, BaaS	Start-ups launch without data centres; scale instantly
Data (big data, analytics)	"Data is the new oil": huge structured and unstructured data	Alternative-data lending, personalisation
Mobile & internet	Smartphones in every pocket (2007–08)	UPI apps, mobile banking, M-PESA
IoT & biometrics	Connected sensors; fingerprints and face	Telematics car insurance, Aadhaar authentication
RPA	Software bots doing repetitive screen tasks	KYC data entry, reconciliations

APIs	Messengers that let apps talk securely	Open banking, payment gateways, Account Aggregator

Close with: no single technology creates FinTech. The **combination** (cloud + mobile + APIs + data + AI) cut the cost of serving a customer, which made small-ticket finance profitable.

🧠 ABCD + MIRA

Also asked as: "Short note on cloud computing and AI" · "Foundational technologies of FinTech" · "Role of big data in finance"

10 marks Short notes: (1) Cloud computing (2) API-based computing vs portals.

Re-exam 2023-24

Cloud computing means renting computing power, storage and software over the internet **on demand, pay-as-you-go**, instead of owning servers (AWS 2006, Azure, Google Cloud).

IaaS
Rent raw servers and storage (AWS EC2).

PaaS
Rent a platform to build apps on (Google App Engine).

SaaS / BaaS
Rent finished software (Salesforce) / banking features via API.

Why it matters for FinTech: no upfront capex, instant scaling (e.g. ZhongAn issues millions of policies on peak shopping days), faster launches. **Risks:** data security, vendor lock-in, data-localisation rules (RBI requires payment data to be stored in India).

	Portal (website for humans)	API (connector for machines)
User	A person logs in and clicks	Another app calls it automatically
Integration	Manual, screen-by-screen	Plug-and-play into any app
Speed & scale	Slow, one-at-a-time	Real-time, millions of calls
Innovation	Only the owner builds features	Third parties build new products (open banking)
Example	Logging into net-banking to download a statement	A loan app fetching your statement via Account Aggregator in seconds

🧠 Portal = a door for people · API = a pipe between machines

2.5–10 marks What are the four pillars of the innovation ecosystem? / Write a note on the current state of FinTech in the world vs India.

2023-24, 2022-23

Four pillars (EY FinTech ecosystem model): Talent · Capital · Policy · Demand.

Talent
Tech + finance skills (IITs, IIMs, ex-bankers).

Capital
Angels, VCs, PE, corporate VC.

Policy
Regulators, sandboxes, licences (RBI, SEBI).

Demand
Customers and businesses willing to adopt.

The course's version adds the actors: start-ups, labs and hubs, incubators and accelerators, regulators and infrastructure (cloud, APIs).

World: over $1.2T invested in FinTech in 2008–22; peak of ~$240B in 2021, then a reset to $164B in 2022 as interest rates rose. The trend moved from disruption to **collaboration and embedded finance**; payments produced the most unicorns.

India: one of the world's largest FinTech hubs, driven by **digital public infrastructure** (India Stack). UPI processed **24.07 billion transactions worth ₹29.37 lakh crore in September 2026 alone**. Strengths: payments and lending at scale, low cost. Challenges: monetisation (zero-MDR UPI), credit risk, and tighter regulation (digital lending rules 2022).

Compare in one line: the West built FinTech on **private** rails (cards, Plaid). India built it on **public** rails (Aadhaar, UPI, Account Aggregator), so adoption is faster and cheaper but margins are thinner.

🧠 **T-C-P-D: "Talented Capital Powers Demand"**

SIMILAR QUESTIONS LIKELY FROM UNIT 1

- ☐ FinTech 1.0 → 2.0 → 3.0 with milestones; four catalysts (Cloud, Smartphones, GFC, Millennials).
- ☐ Apply the FinTech Explained Lens to any company (Paytm, Zerodha, Wise): Who · What · Why · How.
- ☐ Four types of FinTech company and which capabilities each lacks.
- ☐ FinTech for financial inclusion: 1.7B unbanked; M-PESA vs Bitcoin and Wise.

B2 Unit 2 · Financial Services Market Evolution

Asked in 2023-24 and 2024-25 (India-specific framing)

2.5–10 marks **Describe the different types of banking institutions in India.** 2023-24

Framework: who regulates them, and whom they serve. All banks are regulated by the **RBI**.

Type	What they do	Examples
Public sector banks	Government holds the majority; nationwide reach	SBI, Bank of Baroda, PNB
Private sector banks	Privately owned, tech-led	HDFC Bank, ICICI Bank, Axis, Kotak
Foreign banks	Branches or subsidiaries of global banks	Citi (exited retail), HSBC, Standard Chartered
Regional Rural Banks (RRBs)	Credit to farmers and rural areas	Sponsored by public banks
Co-operative banks	Member-owned; urban and rural	Saraswat Bank, state and district co-op banks
Small Finance Banks (2015 licences)	Loans and deposits for the underserved and MSMEs	AU SFB, Equitas, Ujjivan
Payments Banks (2015 licences)	Deposits up to a cap, payments; **cannot lend**	Airtel Payments Bank, India Post Payments Bank
Development Finance Institutions	Long-term project and sector finance	NABARD, SIDBI, EXIM Bank, NaBFID

Also non-bank intermediaries: **NBFCs** (Bajaj Finance), insurers (LIC; regulated by IRDAI), mutual funds and brokers (SEBI), pension funds (PFRDA).

🧠 **"PPF-RCS-PD": Public, Private, Foreign, RRB, Co-op, Small finance, Payments, DFI**

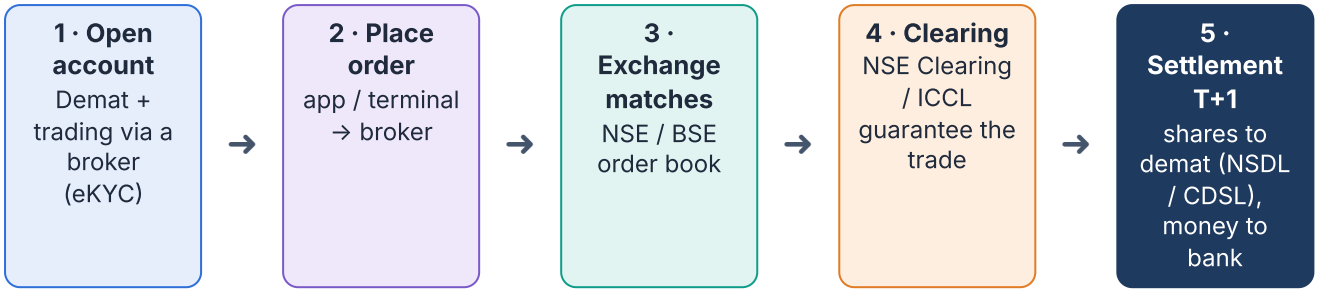
2 marks **"DFIs play an important role in India's economic development." Justify.** 2024-25

Development Finance Institutions give **long-term, low-cost, risky** finance that commercial banks avoid (banks take short-term deposits, so they can't fund 20-year projects). Examples: **NABARD** (agriculture and rural), **SIDBI** (MSMEs), **EXIM Bank** (exports), **NaBFID** (infrastructure, 2021). They fill **market gaps**, support priority sectors and crowd in private capital, which builds jobs and growth.

🧠 **DFIs = long-term money for roads, farms, factories and exporters**

10 marks **Describe the process of equity trading for retail and institutional investors. Highlight**

the role of local participants. What role can the FinTech ecosystem play? 2023-24



Local participant	Role
SEBI	Market regulator; investor protection
Stock exchanges (NSE, BSE)	Match buy and sell orders
Brokers	Route client orders; full-service (ICICI Direct) vs discount (Zerodha, Groww)
Clearing corporations	Act as counterparty, guarantee settlement
Depositories + Depository Participants	Hold shares electronically (NSDL, CDSL)
Custodians (institutional)	Safekeep assets for FPIs and mutual funds
Banks	Fund transfers; ASBA blocks money for IPOs

Retail vs institutional: retail investors trade small quantities through apps and brokers. Institutions (mutual funds, FPIs, insurers) place large orders via **algorithmic and block trades**, use custodians, and are subject to stricter disclosure.

FinTech role: discount brokers with zero-commission delivery (Zerodha: **disruptive** on Pisano's map) · 5-minute eKYC onboarding · fractional and SIP investing · robo-advice · algorithmic and AI trading · UPI mandates for IPOs · better data and analytics. Tie to Unit 2: **transaction costs fall and liquidity rises**.

🧠 **"Open → Order → Match → Clear → Settle (T+1)"**

B3 Unit 3 · AI & ML in Finance

Asked in 2018-19, 2019-20, 2022-23, 2023-24, 2023-24 re-exam

7–10 marks **What is Artificial Intelligence? Difference between supervised and unsupervised learning. Give two daily-life examples.** ★ 3 papers

AI means machines doing tasks that need human-like intelligence: reasoning, learning, understanding language, recognising images. **Machine Learning** is the part of AI where the system **learns rules from data** instead of being programmed with if-then rules.

	Supervised learning	Unsupervised learning
Data	**Labelled** (we know the answer: X → Y)	**Unlabelled** (only X)
Goal	Predict / classify	Find hidden groups and outliers
Methods	Logistic regression, decision trees	K-means clustering, anomaly detection
Finance example	Will this borrower default? Is this a known fraud?	New money-laundering rings; customer segments
Analogy	Studying with an answer key	Sorting a pile of photos without labels

Third type: reinforcement learning learns by reward and penalty (trading bots, rebalancing).

Daily-life AI: Google Maps traffic prediction · Netflix / YouTube recommendations · Siri / Alexa · spam filters · face unlock · bank SMS fraud alerts.

🧠 **S-U-R: Supervised = Scoring · Unsupervised = Unusual · Reinforcement = Rewards**

7–10 marks **How is AI changing the financial sector? Describe its use cases in finance.** ★ 2 papers

Framework: AI across the value chain. Remember **"C-F-C-W-I-R"**:

Credit
Alternative-data scoring; loan decisions in minutes instead of 2–4 weeks (Lendified: SME loan application in under 10 minutes, money in 48 hours).

Fraud & AML
Real-time anomaly detection on card and UPI transactions; unsupervised AML clustering.

Customer service
NLP chatbots (HDFC EVA, SBI SIA); Lemonade's AI Maya sells a policy in about 2 minutes.

Wealth
Robo-advisors: risk profiling, ETF allocation, auto-rebalancing, tax-loss harvesting.

Insurance
Telematics pricing; AI claims (Lemonade's AI Jim pays in about 3 seconds); ZhongAn micro-policies.

Risk & RegTech
Trading algorithms, compliance monitoring, sentiment analysis from news.

Risks (always add for full marks): bias and redlining via proxies (postcode) · the black-box problem vs the duty to explain rejections → use explainable AI (SHAP, LIME) · model drift in crises · privacy.

🧠 **C-F-C-W-I-R + "but explainable"**

Also asked as: "Role of AI in banking" · "AI vs explainability under fair-lending rules" (recommend XGBoost + SHAP) · "BigTech data flywheel"

10 marks **Explain the role a FinTech start-up can play in customer onboarding and auto-insurance.** 2023-24

A · Customer onboarding (digital KYC)



Benefits: onboarding falls from **days and paper to minutes**; cost per customer drops sharply; fewer errors and frauds; reach into rural India. Examples: eKYC APIs (Signzy, IDfy), DigiLocker document pull.

B · Auto-insurance (InsurTech)

- **Distribution:** compare and buy online in minutes (PolicyBazaar, Acko, Digit).
- **Pricing: telematics / usage-based insurance** ("pay-as-you-drive"): sensors track speed and braking, so safe drivers pay less (IRDAI has allowed add-on PAYD covers).
- **Claims:** upload photos or video; computer vision estimates damage; fast or instant settlement; AI fraud checks (as Lemonade's AI Jim does).
- **Engagement:** renewal nudges, driving scores, embedded insurance at car purchase.

🧠 **Onboarding = "Verify → Read → Match → Check → Go" · Insurance = "Buy · Price · Claim · Engage"**

7–10 marks **What is Robotic Process Automation (RPA)? Explain.** ★ 2 papers · not in slides

RPA uses **software "bots"** that copy what a human does on a computer screen (click, copy, paste, type, read emails) to automate **repetitive, rule-based, high-volume** tasks. There are no physical robots.

	RPA	AI
Works on	Fixed rules ("if X then do Y")	Patterns learned from data
Thinks?	No, it just follows steps	Yes, it predicts and decides
Best for	Data entry, reconciliation	Credit scoring, fraud detection

Banking uses: KYC and account-opening data entry · loan-application processing · daily account reconciliation · report generation for regulators · credit-card dispute handling · updating customer addresses across systems.

Benefits: 24×7, fast, error-free, cheap, audit trail, staff freed for judgement work. **Limits:** breaks when screens change; can't handle exceptions; needs governance. Tools: UiPath, Automation Anywhere, Blue Prism. **RPA + AI = "intelligent automation".**

🧠 **RPA = a digital clerk that never sleeps but never thinks**

B4 Unit 4 · Blockchain & Crypto

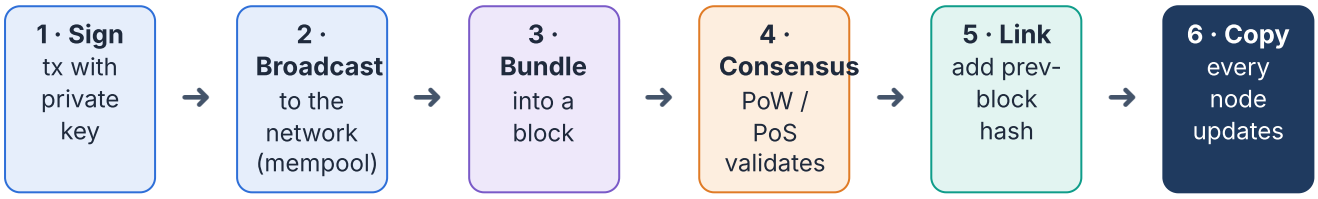
Asked in 2018-19, 2019-20, 2022-23, 2023-24, 2023-24 re-exam, 2024-25

10 marks What is blockchain technology? Explain how it works and how it applies to FinTech.

★ 4 papers

Definition: a blockchain is a **shared, append-only digital ledger** copied across many computers (nodes). Transactions are grouped into **blocks**, and each block holds the **hash of the previous block**, so the chain can't be secretly changed. It solves the **double-spend problem without a middleman** (Bitcoin, 2008–09).

How it works: 6 steps



Key features (remember "D-I-T-S"): **D**ecentralised · **I**mmutable · **T**ransparent · **S**ecure (cryptography + consensus). Plus smart contracts on Ethereum ("if A then pay B").

Types: public / permissionless (Bitcoin, Ethereum) · private / permissioned (Hyperledger, R3 Corda, used by banks) · consortium.

FinTech applications

Use case	How blockchain helps
Cross-border payments	Ripple / XRP settles in seconds instead of 3–5 days via SWIFT
Trade finance	Digital letters of credit and bills of lading; one shared copy for all parties
KYC sharing	Banks verify a customer once and share the record
Securities settlement and tokenisation	Faster clearing; fractional bonds and property
Crypto, DeFi, stablecoins, CBDC	Digital money, lending pools, e-Rupee pilots (Dec 2022)
Insurance and land records	Smart-contract claims; tamper-proof titles

Limits: slow (Bitcoin 2–4 TPS vs Visa 24,000), PoW energy (~73 TWh), regulation, failures of centralised exchanges (FTX).

🧠 **"Sign · Broadcast · Bundle · Agree · Link · Copy" + D-I-T-S**

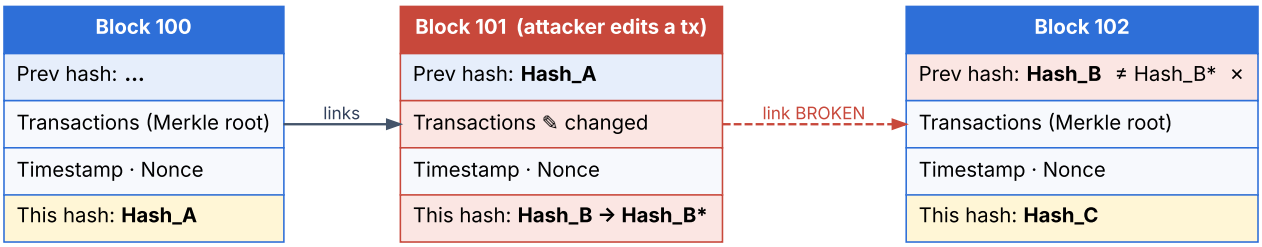
Also asked as: "Explain the two key use cases of blockchain in FinTech" (pick cross-border payments + trade finance or KYC; give a how + an example for each) · "Define blockchain and list a use case"

7 marks How is blockchain used for fraud reduction?

2019-20

1. **Immutability:** a recorded entry can't be edited. Change one block and its hash breaks every later link (needs a 51% attack).

2. **Single shared truth:** all parties see the same ledger, so no fake duplicates (e.g. the **same invoice financed twice** in trade finance).
3. **Digital signatures:** only the private-key owner can authorise a transfer, which proves who did what.
4. **Full audit trail:** every transaction is time-stamped and traceable (anti-money laundering).
5. **Smart contracts:** money is released only when conditions are met, so no manual tampering.
6. **Identity / KYC on chain:** verified identities cut synthetic-identity fraud.



🧠 **"Can't edit · Everyone sees · Signed · Traced · Auto-executed · Verified"**

2.5 marks **Explain "hashing" in blockchain.** 2023-24

Hashing turns **any input into a fixed-length "fingerprint"**; SHA-256 gives 256 bits, shown as 64 hex characters. It is **one-way** (you can't get the input back), **deterministic** (same input, same hash), shows the **avalanche effect** (one letter changed gives a totally different hash: "fintech" → 110599cc..., "Fintech" → 2dd47261...) and is **collision-resistant**. In blockchain, each block stores the previous block's hash (linking and tamper-proofing), and miners search for a nonce that makes the hash smaller than the target (Proof-of-Work).

🧠 **F-DOAC: Fixed · Deterministic · One-way · Avalanche · Collision-resistant**

2.5–12 marks **What is a digital signature? Explain 3 differences between digital and paper signatures. Define private and public key.** ★ 2 papers

Public-key (asymmetric) cryptography gives every user a key pair. The **private key** is secret, kept only by the owner and used to **sign**. The **public key** is shared openly and used by anyone to **verify** the signature (or to encrypt a message only the private key can open).

Digital signature: an electronic code made by **hashing** a document and **encrypting the hash with the signer's private key**. The receiver decrypts it with the public key and compares hashes. A match proves **who signed** (authentication), that the document **wasn't changed** (integrity), and that the signer **can't deny it** (non-repudiation). Legal in India under the **IT Act, 2000**; e.g. Aadhaar **eSign**.



Basis	Paper signature	Digital signature
Security	Easy to forge or copy	Cryptographic, practically impossible to forge

Tamper detection	Can't show if the page was changed later	Any change breaks the hash, so it's detected instantly
Verification	Manual, by eye or an expert	Automatic, by software, anywhere
Speed and cost	Courier, print, store	Instant, paperless
Uniqueness	Same signature on every document	Different for every document (depends on its hash)

🧠 **Private = my pen (sign) · Public = everyone's magnifying glass (verify) · A-I-N: Authentication, Integrity, Non-repudiation**

7–10 marks **What is trade finance? How will FinTech disrupt it?** ★ 2 papers · not in slides

Trade finance is the set of products that help importers and exporters trade safely: the exporter wants to be paid, the importer wants the goods. Banks bridge that **trust and cash gap**.

Instruments: Letter of Credit (LC: bank guarantees payment once documents are presented) · bank guarantees · invoice factoring / bill discounting · supply-chain finance · export credit · documentary collections.

Pain points: very **paper-heavy** (bills of lading, invoices couriered between banks) · slow (days to weeks) · **fraud** (fake or double-financed invoices) · high cost, so **MSMEs are left out** (a large trade-finance gap).

FinTech solution	Effect
Blockchain platforms (digital LC, electronic bill of lading)	One shared, tamper-proof record; settlement in hours
Invoice-discounting platforms, e.g. India's **TReDS** (RXIL, M1xchange, Invoicemart)	MSMEs sell invoices of large buyers to financiers online
AI / OCR document checking	Automatic compliance and sanctions checks
Alternative-data credit scoring (GST data, Account Aggregator)	Loans for MSMEs without collateral
APIs + cross-border payment rails	Faster, cheaper FX and payments (Wise-style)
Smart contracts + IoT tracking	Auto-payment when the shipment arrives

🧠 **Trade finance problem = "Paper, Slow, Fraud, MSME gap" → FinTech = "Digitise, Share, Score, Automate"**

B5 Unit 5 · Payments & Digital Wallets

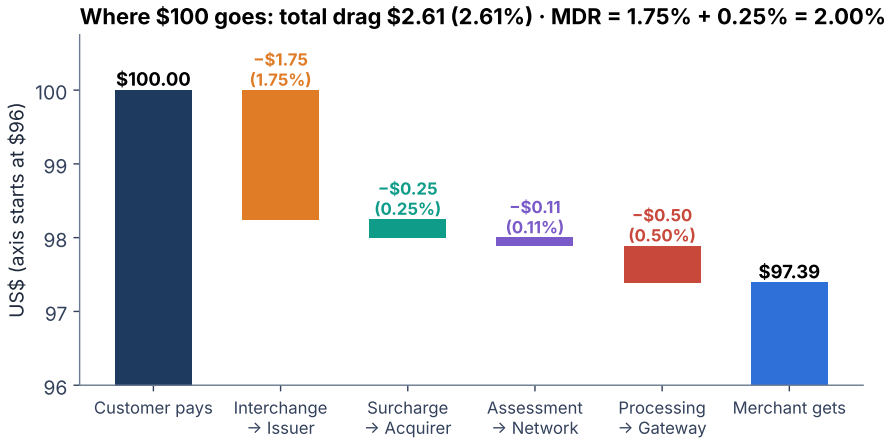
Asked in 2022-23, 2023-24, 2023-24 re-exam (plus the India Stack "cashless" layer)

10 marks **Explain the working of a credit-card-based payment system.** 2023-24

A card payment is a **four-party scheme**: cardholder, merchant, acquirer (merchant's bank), issuer (cardholder's bank), linked by a **card network** (Visa, Mastercard, RuPay).

① **Cardholder** taps / swipes → ② **Merchant** POS / gateway → ③ **Acquirer** forwards → ④ **Network** routes → ⑤ **Issuer** checks limit, fraud → approves

Step 1 · Authorisation (seconds): the request goes merchant → acquirer → network → issuer; the approval comes back the same way. **Step 2 · Clearing and settlement (T+1 / 1–2 days, batched):** the issuer pays the acquirer, which credits the merchant after **deducting fees**. **Step 3 · Billing:** the issuer bills the cardholder monthly, interest-free if paid in full by the due date.



Fees (King's $100 example): interchange $1.75 → issuer · acquirer surcharge $0.25 (MDR = 2.00%) · network $0.11 · gateway $0.50 → **merchant gets $97.39**. The issuer earns the most because it carries credit risk, fraud and rewards. India: **tokenisation is mandatory since Oct 2022** (merchants can't store card numbers); **OTP / 2-factor authentication**; RuPay credit cards can be used on UPI.

🧠 **"Tap → Ask → Approve → Settle → Bill" · M-A-I-C**

10 marks **Explain acquiring bank, issuing bank and the "network". What are the various networks in India?** Re-exam 2023-24

Issuing bank

The cardholder's bank. Issues the card, approves transactions, bears default and fraud risk, earns **interchange**. E.g. HDFC Bank issuing your card.

Acquiring bank

The merchant's bank. Signs up merchants, provides POS / gateway, settles money to the merchant, earns a margin. E.g. Axis Bank or Pine Labs for a store.

Network (scheme)

The rails and rulebook linking issuers and acquirers worldwide; routes and clears messages, sets interchange, earns assessment fees. **Doesn't hold the customer's money.**

Networks in India	Type
Visa, Mastercard	Global four-party card networks
RuPay (NPCI, 2012)	India's domestic card network: lower cost, data kept in India, Jan Dhan cards
American Express, Diners Club	Three-party (closed) networks: issuer and acquirer in one
NPCI rails (NPCI set up 2008 by RBI and IBA)	**UPI** (2016), **IMPS** (24×7 instant transfers), NACH (bulk mandates, salaries), AePS (Aadhaar-enabled payments), BBPS (bill pay), NETC **FASTag** (tolls)
RBI-run systems	**NEFT** (24×7 since Dec 2019) and **RTGS** (high-value, real-time)

🧠 **Issuer = "your" bank · Acquirer = "shop's" bank · Network = the road between them**

2.5–10 marks **Why are digital payments the most popular segment in the Indian start-up ecosystem? / Short note on UPI.** ★ 2 papers

UPI (Unified Payments Interface) was launched by **NPCI in 2016**. It is a real-time, 24×7, bank-to-bank (A2A) payment system using a **virtual payment address** (name@bank) or a QR code. One app can link many bank accounts, and the PIN is held by the bank. P2P and P2M payments are **zero-MDR** for users. **Scale: 24.07 billion transactions worth ₹29.37 lakh crore in September 2026.** Built on India Stack's cashless layer; interoperable (PhonePe, Google Pay, Paytm, BHIM). Features: AutoPay mandates, UPI Lite, credit on UPI (RuPay credit cards), international links (Singapore, UAE and others).

Why payments are start-ups' favourite (use the Lens):

1. **Huge demand:** a billion phones, cheap data (post-2016), demonetisation (Nov 2016) and COVID pushed cash users online.
2. **Public rails are free to build on:** UPI, Aadhaar and eKYC cut onboarding and infrastructure costs (no need to build card networks).
3. **High frequency = engagement:** people pay daily, so payments are the best customer-acquisition hook (the subsidy side of a platform).
4. **Gateway to cross-sell:** payments data feeds lending, insurance and wealth (the money side): Paytm, PhonePe, BharatPe.
5. **Policy support:** Digital India, RBI and NPCI innovation, government incentives for RuPay / UPI.

Catch: zero MDR means thin margins, so most payment start-ups lose money (Paytm's processing costs ate 55–69% of revenue) and must monetise via **lending, devices (Soundbox), merchant SaaS**.

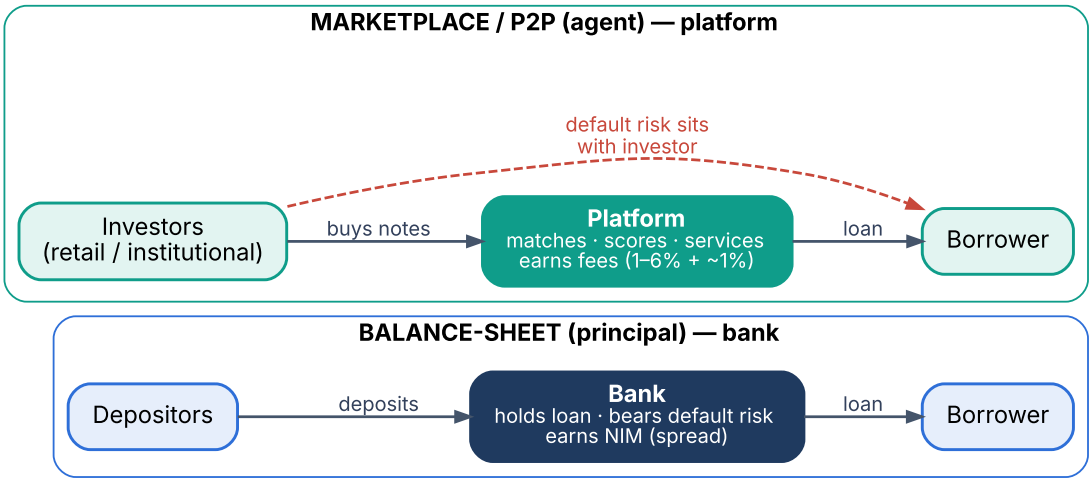
🧠 **"Phones + Free rails + Frequency + Cross-sell + Policy"**

B6 Unit 6 · P2P Lending, Crowdfunding & Digital Lending

Asked in 2019-20, 2023-24 re-exam, 2024-25

10 marks **What is P2P lending? Mention the different types of lending platforms provided by FinTech. Also give the types of credit systems.** 2024-25

P2P lending: an online **marketplace** that matches **borrowers directly with lenders / investors**. The platform scores the borrower, sets the rate, collects repayments and earns **fees** (origination 1–6% + servicing ~1%). **Default risk sits with the lender**, not the platform (an agent model, not a principal model). India: RBI regulates P2P platforms as **NBFC-P2P** (2017 directions). Limits: a lender can have at most **₹50 lakh** across all platforms and a borrower at most **₹10 lakh**. Examples: Faircent, LenDenClub; globally LendingClub, Funding Circle.



Types of FinTech lending platforms	How it works	Example
P2P marketplace	Matches individuals; platform earns fees	Faircent, LendingClub
Balance-sheet (NBFC) digital lender	Lends its own money via an app	KreditBee, Lendingkart
Bank / NBFC partnership (co-lending, Lending Service Provider)	FinTech sources and scores; a regulated entity lends (RBI digital lending rules 2022)	Many app-based loans
Buy Now Pay Later / embedded credit	Credit at checkout	Simpl, LazyPay, Amazon Pay Later
Invoice / supply-chain finance	Advances against invoices	TReDS platforms
Crowdfunding (debt / equity)	Many small investors fund one project	Kickstarter (rewards), Ketto (donation)

Types of credit systems (two ways to answer):

- **By how credit is assessed:** ① **traditional bureau-based scoring** (CIBIL, Experian, Equifax, CRIF High Mark: score 300–900 from repayment history) ② **alternative-data / AI scoring** (bank statements via Account Aggregator, GST, UPI, utility bills) for thin-file borrowers ③ **relationship / collateral-based** lending (gold, property).

- **By type of credit:** secured vs unsecured · revolving (credit card, credit line) vs instalment (EMI loans) · short-term (working capital, BNPL) vs long-term (home loans).

🧠 **P2P = "matchmaker, not money-holder" · Credit systems = "Bureau, Big data, Backed by collateral"**

7 marks **What is crowdfunding? What is the future of crowdfunding in India?** 2019-20

Crowdfunding raises money from **many people in small amounts via an online platform**, instead of one bank or investor.

Donation	Reward	Equity	Debt / P2P
Nothing in return. Ketto, Milaap, GoFundMe.	Product or perk. Kickstarter (all-or-nothing; 5% fee), Wishberry.	Shares in a start-up (strictly limited in India).	Loans with interest. Faircent. ~90% of global volume.

Future in India: ✔ **donation crowdfunding** is booming (medical, social causes) · ✔ **P2P** is growing under the RBI NBFC-P2P framework · ✔ UPI makes contributing instant · ✔ **Social Stock Exchange** (SEBI, 2022) lets non-profits raise funds · ✖ **equity crowdfunding** to the public is **not permitted** by SEBI (only private placements or angel networks), so it stays limited · ✖ fraud risk and trust (Ezubao lesson from China). **Outlook:** grows through donation, P2P and community finance; equity only if SEBI creates a framework like the US JOBS Act Title III.

🧠 **D-R-E-D · India: "Donation booms, Debt regulated, Equity blocked"**

2.5 marks **Why do most FinTech start-ups concentrate on digital lending?** Re-exam 2023-24

Because **lending is where the money is**: payments earn tiny or zero margins, but loans earn **interest and fees**. India has a **huge unmet credit demand** (MSMEs and thin-file consumers ignored by banks). India Stack lowers costs (eKYC onboarding, Account Aggregator and UPI data for underwriting, eSign / eNACH for paperless documents and repayments), and payment apps already hold the customer data to cross-sell credit. **Risks:** high defaults, predatory "loan apps", hence RBI's **2022 digital-lending guidelines** (loans must be disbursed and repaid directly via regulated entities, with full fee disclosure).

🧠 **"Margin + Demand + Data + Cheap rails"**

B7 Unit 7 · Wealth Management, Robo-Advisors & Open Banking

Asked in 2023-24 re-exam, 2024-25 (+ APIs / Account Aggregator in India Stack questions)

10 marks **How have FinTech companies transformed the wealth-management business model, and what strategies do they use to personalise services? Explain with examples. / Role of a FinTech start-up in mutual funds and wealth management.** ★ 2 papers

Old model vs FinTech model

	Traditional	FinTech
Fees	1–2% of assets (or high commissions on regular MF plans)	0.25–0.50% robo fees, or **direct MF plans** with no commission
Minimum	₹ lakhs / $250k+	SIP from ₹100–500; $500–5,000
Onboarding	Paper, branch visits	eKYC in minutes
Advice	Human relationship manager	Algorithms + human (hybrid)
Who it serves	The rich	Mass market, young earners, HENRYs

How they personalise: the robo workflow

Risk questionnaire → **Goal-based plan** retirement, house → **Auto portfolio** ETFs / index funds → **Auto-rebalance** + tax-loss harvesting → **Nudges** step-up SIPs, alerts

Strategies: ① goal-based investing ② AI risk profiling and behavioural nudges ③ direct mutual funds at zero commission (Groww, Kuvera, Zerodha Coin) ④ **smallcases** / thematic baskets ⑤ fractional and global investing ⑥ hybrid "high-tech, high-touch" advice ⑦ **Account Aggregator** data for a full financial picture ⑧ education content and community.

FinTech roles in mutual funds specifically: digital onboarding (eKYC, FATCA) · SIP automation via UPI AutoPay / eNACH · comparison and analytics (returns, expense ratio, risk) · direct-plan distribution · portfolio tracking across AMCs (CAMS / KFintech, MF Central) · B2B tech for AMCs and distributors (e.g. MF Utilities, BSE StAR MF).

Examples: Wealthfront and Wealthsimple (35–50 bps vs 145 bps, minimum $5k vs $250k) · **Vanguard** launched its own robo in 2015 and won over 50% of robo assets in 2 years · India: Zerodha, Groww, Kuvera, Scripbox, INDmoney.

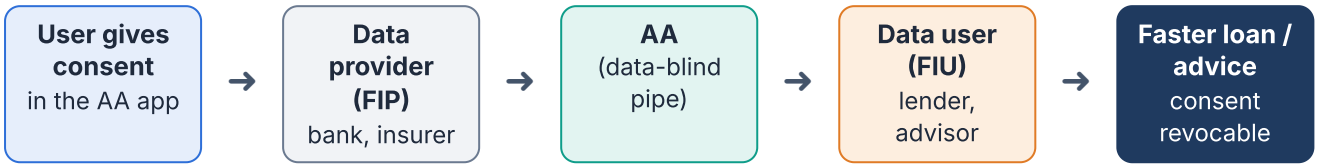
Challenges: high customer-acquisition cost and thin margins, which pushes them toward hybrid and B2B models; behaviour in market crashes; SEBI rules separating advice from distribution (Registered Investment Advisers).

🧠 **"Cheaper · Smaller tickets · Faster onboarding · Smarter (algorithms) · Wider reach"**

7–10 marks **What is open banking? Explain APIs and India's Account Aggregator framework.**

Likely (Unit 7 + India Stack consent layer)

Open banking lets **customers own their financial data** and share it, with consent, with third-party apps through secure **APIs**. EU: **PSD2** (2016); UK Open Banking (2016). India's version is the **Account Aggregator (AA)**, an RBI-licensed NBFC-AA (framework 2016, live Sept 2021). The AA is a **consent manager** that moves encrypted data from a bank or insurer (Financial Information Provider) to a lender or advisor (Financial Information User). It is **data-blind**: it can't read or store the data.



Three types of API: closed (internal) · partner · open (public). **BaaS** layers bottom-up: **L-M-D-U** (Licensed bank → Middleware → Distributor app → User).

🧠 **Open banking = "My data, my consent, any app"**

C India-specific & later-unit questions

Outside Units 1–7 as taught, but asked again and again. Prepare C1–C3 for sure.

INDIA STACK = 6/6 PAPERS

10–14 marks **C1 · What is India Stack? What are its components / layers (with use cases)? What benefits does it give citizens, and what role does it play in the FinTech world?** ★★ 6 / 6 papers

Definition: India Stack is a set of **open, government-backed digital public infrastructure (DPI)**: APIs that let any company verify identity, collect documents, move money and share data **without physical presence, paper or cash**. It is the reason Indian FinTech scaled so cheaply. It started with Aadhaar (UIDAI, 2009).

4 · CONSENT	**Consent layer**: **DEPA** (Data Empowerment and Protection Architecture) + **Account Aggregator** (live 2021). *Use: share bank statements with a lender in seconds, with your permission.*
3 · CASHLESS	**Cashless layer**: **UPI** (2016), IMPS, **AePS** (Aadhaar-enabled payments), Aadhaar Payments Bridge (DBT). *Use: pay any shop by QR; subsidies straight into bank accounts.*
2 · PAPERLESS	**Paperless layer**: **Aadhaar eKYC**, **eSign**, **DigiLocker**. *Use: open a bank or demat account and sign forms without paper.*
1 · PRESENCE-LESS	**Presence-less layer**: **Aadhaar** biometric / OTP authentication (over 130 crore residents). *Use: prove "I am who I say I am" from anywhere.*

MNEMONIC · BOTTOM-UP "P-P-C-C"

P resence-less (who are you?) | **P** aperless (show documents) | **C** ashless (move money) | **C** onsent (share data)

"**P**eople **P**ay **C**ash**C**ash-free". Identity → Documents → Money → Data.

Benefits to citizens

Financial inclusion
Jan Dhan + Aadhaar + mobile (JAM) opened over 50 crore basic accounts; anyone can bank.

Direct Benefit Transfer
Subsidies paid straight into accounts, with fewer middlemen and less leakage.

Cheap, instant payments
UPI: free, 24×7. 24.07 billion transactions in September 2026.

Speed & convenience
KYC in minutes, documents in DigiLocker, eSign from home.

Access to credit
Account Aggregator and UPI data let thin-file borrowers get loans.

Control of data
Consent-based sharing, revocable anytime.

Role in the FinTech world

It acts as **public rails**, so start-ups don't have to build identity, KYC or payment systems themselves. That cuts **customer-acquisition and onboarding costs** and lets them serve small-ticket customers profitably: Paytm, PhonePe and Google Pay on UPI, Zerodha and Groww on eKYC, digital lenders on Account Aggregator. It is copied globally (UPI links abroad; the "India Stack" model offered to other countries). Newer additions: **ONDC** (commerce), **OCEN** (credit), ABDM (health).

Concerns (for balance): privacy and surveillance risk, Aadhaar authentication failures excluding people, cyber fraud, concentration on UPI apps.

🧠 **India Stack = "Identity · Documents · Money · Data, all as APIs"**

Also asked as: "4 layers of the Aadhaar stack" · "What is JAM and two uses" · "Benefits of India Stack to citizens" · "Explain the cashless layer" (see C2)

10 marks **C2 · Explain the "cashless" layer of India Stack. How did it transform payments in India?**

Re-exam 2023-24

The cashless layer is a set of **interoperable, real-time payment rails** run mainly by **NPCI**: **UPI** (any bank → any bank by VPA / QR), **IMPS** (24×7 instant transfers), **AePS** (withdraw or transfer using Aadhaar + fingerprint at a micro-ATM, which serves rural users), **Aadhaar Payments Bridge** (government DBT to Aadhaar-linked accounts), plus BHIM, UPI 123Pay (feature phones) and UPI Lite.

Before	After the cashless layer
Cash-heavy; cards only in cities (MDR, POS terminals)	QR codes everywhere, from tea stalls to malls, at zero MDR
Transfers needed account number + IFSC, NEFT in batches	Instant 24×7 with a phone number or VPA
Each wallet a closed loop	**Interoperable**: any app, any bank
Subsidies leaked through middlemen	DBT straight to accounts

Transformation in numbers: UPI went from launch in 2016 to **24.07 billion transactions (₹29.37 lakh crore) in September 2026**, with daily volumes above 800 million. India leads the world in real-time payments. It enabled new business models (Soundbox, UPI AutoPay SIPs, credit on UPI) and accelerated after demonetisation (2016) and COVID.

🧠 **"Any app · Any bank · Any time · Zero cost"**

4–10 marks **C3 · Short notes: Aadhaar eKYC · JAM trinity**

★ 2018-19, 2022-23

Aadhaar eKYC: instead of photocopies, a bank or FinTech asks UIDAI to verify the customer using **OTP or biometrics**. UIDAI returns the verified name, address, photo and date of birth **digitally**. Benefits: KYC in minutes, near-zero cost, no forged documents, paperless, works in rural areas. Rules: needs customer consent; offline / masked-Aadhaar and video KYC (RBI 2020) are alternatives; privacy safeguards after the Supreme Court's 2018 Aadhaar judgment.

JAM = Jan Dhan + Aadhaar + Mobile. Jan Dhan (Aug 2014) gave every household a zero-balance bank account; **Aadhaar** gives each person a unique digital identity; **Mobile** connects them for instant access. Two uses: ① **Direct Benefit Transfer** (LPG subsidy, scholarships, PM-KISAN straight into accounts, cutting leakages) ② **financial inclusion** (micro-insurance and pensions via Jan Dhan accounts, UPI payments, small loans).

🧠 JAM = "Account + Identity + Phone = money reaches the right person directly"

2–10 marks **C4 · What initiatives has the Indian Government / RBI taken to promote FinTech?** 2024-25 (twice)

Government of India	RBI-pioneered
• **Digital India** (2015): connectivity and e-services • **Startup India** (2016): tax breaks, funding support • **JAM trinity** and Pradhan Mantri Jan Dhan Yojana (2014) • **Aadhaar** (UIDAI 2009; Aadhaar Act 2016) • **BHIM app** (2016), e-RUPI vouchers (2021) • **IFSCA at GIFT City** (2020): FinTech sandbox and incentive scheme • **DPDP Act 2023**: data-protection clarity • **ONDC, OCEN**, GSTN data for credit	• Set up **NPCI** (2008, with IBA) → **UPI, IMPS, RuPay**, AePS, BBPS • 24×7 NEFT / RTGS; **Payments Banks** and **Small Finance Banks** licences (2015) • **Regulatory sandbox** (2019) • **Account Aggregator** framework (live 2021) • **Video KYC** (2020); card **tokenisation** (2022) • **Digital lending guidelines** (2022) • **e-Rupee CBDC** pilots (wholesale Nov 2022, retail Dec 2022) • **TReDS** for MSME invoices; **RBI Innovation Hub** and a FinTech Department (2022) • Payments Infrastructure Development Fund; UPI 123Pay for feature phones

🧠 Govt = "Digital India, Startup India, JAM, IFSCA" · RBI = "NPCI rails, new banks, sandbox, AA, CBDC"

7 marks **C5 · What is a regulatory sandbox? Explain the sandbox for FinTech in India.** 2019-20

A **regulatory sandbox** lets FinTechs **test new products live with real customers**, on a **limited scale and for a limited time**, under **relaxed rules and close regulator supervision**. Innovation gets tested safely before full regulation.

Apply eligible innovation → **Regulator selects** a cohort → **Test** limited users, time, value → **Monitor** reports, safeguards → **Exit** approve / modify / stop

India: RBI's **Enabling Framework for Regulatory Sandbox (2019)**. Themed cohorts: retail payments (1st), cross-border payments, MSME lending, prevention and mitigation of financial frauds. SEBI, IRDAI and IFSCA run their own sandboxes. **Benefits:** faster time to market, regulators learn early, consumers are protected. **Limits:** small scale, slow approvals, no guarantee of a final licence.

🧠 Sandbox = "test in a fenced playground with the referee watching"

2–7 marks **C6 · What is regulation and how are rules enforced? What are the regulatory problems around FinTech? Name apex bodies and the year they were set up.** ★ 2018-19, 2019-20, 2024-25

Regulation = rules made by an authorised body to protect **consumers**, ensure **financial stability** and keep **markets fair**. **Enforcement:** ① licensing / registration ② ongoing reporting and audits ③ inspections ④ penalties and fines ⑤ business restrictions or licence cancellation (e.g. RBI barred Paytm Payments Bank from new deposits in 2024) ⑥ consumer grievance systems (RBI Ombudsman).

Regulatory problems around FinTech ("GRAPES"): **G**rey zones and regulatory arbitrage (activity-based, not entity-based) · **R**egulator overlap (RBI, SEBI, IRDAI, MeitY) · **A**ML / KYC gaps and fraud · **P**rivacy and data misuse · **E**xploitative practices (predatory loan apps, mis-selling) · **S**peed mismatch (tech moves faster than rules) and **S**ystemic / cyber risk. Crypto still has no clear framework (taxed at 30% + 1% TDS since 2022).

Apex body	Regulates	Established
RBI	Banks, NBFCs, payments, monetary policy	1935 (1 April)
SEBI	Securities markets, mutual funds, brokers	1988; statutory powers 1992
IRDAI	Insurance	IRDA Act 1999; set up 2000
PFRDA	Pensions (NPS)	2003; statutory 2014
NABARD	Rural and agricultural finance (DFI)	1982
NPCI	Retail payment rails (UPI, RuPay)	2008

🧠 **Problems = GRAPES · Bodies by year: "RBI '35 → NABARD '82 → SEBI '88/'92 → IRDA '99 → PFRDA '03 → NPCI '08"**

10 marks **C7 · What is privacy and why is it important? What are the types of cyber threats? / Describe data security and privacy requirements of the financial sector and the technologies used.** ★ 2 papers

Privacy is a person's right to **control how their personal and financial data is collected, used and shared**. In finance it matters because data reveals income, spending and identity. Misuse leads to fraud, identity theft, discrimination and loss of trust. India: **Right to Privacy** is a fundamental right (Supreme Court, 2017) and the **DPDP Act 2023** (consent, purpose limitation, penalties).

Cyber threat	Simple meaning
Phishing / vishing / smishing	Fake emails, calls or SMS that steal OTPs and passwords
Malware & ransomware	Bad software that steals data or locks systems for ransom
Identity theft / SIM swap / account takeover	A criminal pretends to be you
Data breach & insider threat	Customer data leaked by hackers or employees
DDoS	A flood of traffic crashes banking or UPI services
Man-in-the-middle / API attacks	Intercepting data between app and server
Deepfakes & social engineering	AI-faked voices or video to trick staff or customers

Requirements: RBI cyber-security framework for banks (2016) and IT governance rules · **data localisation** of payment data in India (RBI 2018) · **CERT-In** incident reporting within 6 hours (2022) · KYC / AML · PCI-DSS for card data · ISO 27001 · DPDP consent and breach notification · card tokenisation (2022).

Technologies: encryption (data at rest and in transit) · **tokenisation** · multi-factor and biometric authentication · firewalls, intrusion detection, SOC / SIEM monitoring · **AI fraud detection** · zero-trust access · hardware security modules · blockchain audit trails · regular penetration testing and staff training.

🧠 Protect = "Encrypt · Tokenise · Authenticate · Monitor · Train"

2–12 marks **C8 · Name the 5 stages of Design Thinking (one line each). / Explain the Minimum Viable Product (MVP).** ★★ 4 papers



Design Thinking (Stanford d.school / IDEO) is a **human-centred, iterative** way to solve problems. It is non-linear: you loop back after testing. **FinTech example:** UPI-style payments for street vendors. Empathise (vendors hate cash and change) → Define ("accept money without a POS machine") → Ideate (QR, voice alerts) → Prototype (printed QR + Soundbox) → Test (vendor feedback) → scale. The link to Unit 1: the FinTech Lens is Empathise + Define (Who? What?).

MVP: the **simplest version of a product** with just enough features to test whether customers want it, using **Build → Measure → Learn** (Eric Ries, *The Lean Startup*). It saves money and time and gets real feedback early. **Examples:** Wealthsimple began as an **Excel ETF portfolio for friends** (1,000 clients in months); Wise began as two founders swapping EUR and GBP with each other.

🧠 "Every Dog Is Pretty Tame": Empathise · Define · Ideate · Prototype · Test · MVP = "smallest thing that proves demand"

2.5–12 marks **C9 · What is a ULIP and which regulator governs it? / What is monetary policy and its objectives?** 2023-24 re-exam, 2018-19

ULIP (Unit Linked Insurance Plan) = **life insurance + market investment in one product**. Part of the premium buys life cover; the rest is invested in equity, debt or balanced funds of your choice (NAV-based, you can switch funds). It has a **5-year lock-in** and tax benefits (subject to limits). **Regulator: IRDAI**, not SEBI. The 2010 SEBI–IRDAI turf war was settled in IRDAI's favour. FinTech angle: online comparison and purchase (PolicyBazaar), lower charges.

Monetary policy = RBI's management of **money supply and interest rates**. **Objective: price stability** (inflation target **4% ± 2%**) while supporting **growth**; also financial stability and exchange-rate stability. Decided by the **Monetary Policy Committee** (6 members, since 2016). **Tools:** repo rate, Standing Deposit Facility / reverse repo, CRR, SLR, open market operations. FinTech link: faster transmission through digital payments, and the e-Rupee (CBDC).

🧠 ULIP = "Insure + Invest, IRDAI's baby" · Monetary policy = "Repo, CRR, SLR, OMO → control inflation 4% ± 2%"

C10 · EXPAND THESE (ASKED AS A 12-MARK QUESTION IN 2018-19)

NBFC Non-Banking Financial Company
P2P Peer-to-Peer
MoF Ministry of Finance
RBI Reserve Bank of India
API Application Programming Interface
GSTN Goods and Services Tax Network
SEBI Securities and Exchange Board of India
UPI Unified Payments Interface
IRDA(I) Insurance Regulatory and Development Authority (of India)
IoT Internet of Things
DLT Distributed Ledger Technology
ML Machine Learning
NPCI National Payments Corporation of India
JAM Jan Dhan–Aadhaar–Mobile
KYC / eKYC (Electronic) Know Your Customer
AML Anti-Money Laundering
IMPS Immediate Payment Service
NEFT / RTGS National Electronic Funds Transfer / Real Time Gross Settlement
AePS Aadhaar-enabled Payment System
NACH National Automated Clearing House
BBPS Bharat Bill Payment System
DBT Direct Benefit Transfer
DEPA Data Empowerment and Protection Architecture
AA Account Aggregator
CBDC Central Bank Digital Currency (e₹)
TReDS Trade Receivables Discounting System
PFRDA Pension Fund Regulatory and Development Authority
NABARD / SIDBI National Bank for Agriculture and Rural Development / Small Industries Development Bank of India
IFSCA International Financial Services Centres Authority
RPA · MVP · ULIP · MDR Robotic Process Automation · Minimum Viable Product · Unit Linked Insurance Plan · Merchant Discount Rate

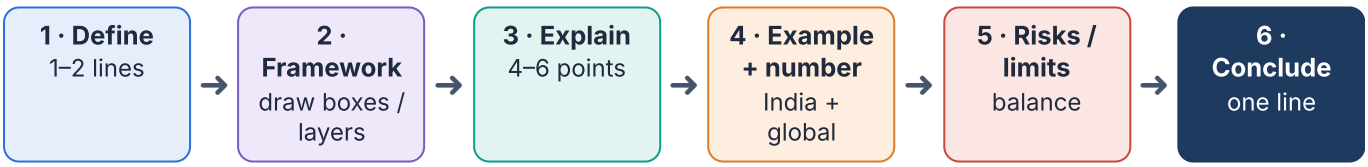
D Last-minute answer kit

If you only have 30 minutes before the exam

Top 10 questions to prepare (by past frequency and fit with Units 1–7)

#	Question	Section	Hook
1	India Stack: layers, components, benefits, role	C1	P-P-C-C
2	Blockchain: what, how, FinTech use cases, fraud reduction	B4	Sign-Broadcast-Bundle-Agree-Link-Copy · D-I-T-S
3	What is FinTech + how it disrupts banking	B1	4 value words + I-T-L-R-T
4	AI: supervised vs unsupervised + AI in finance	B3	S-U-R · C-F-C-W-I-R
5	Technologies driving FinTech	B1	ABCD + MIRA
6	Credit card system / issuer, acquirer, network / networks in India	B5	Tap-Ask-Approve-Settle-Bill · $97.39
7	Design Thinking + RPA + MVP	C8, B3	"Every Dog Is Pretty Tame"
8	Wealth management transformation / MFs	B7	Robo workflow
9	P2P lending, platform types, credit systems; crowdfunding in India	B6	Matchmaker, not money-holder · D-R-E-D
10	Trade finance and FinTech	B4	Paper-Slow-Fraud-MSME gap

The universal 10-mark answer template



2-mark one-liners (Q1 short answers)

FinTech	Technology delivering financial services cheaper, easier, faster and more conveniently than traditional banks.
Blockchain	Shared, tamper-proof digital ledger of transactions linked by hashes and validated by consensus.
Hashing	One-way, fixed-length fingerprint of data; any change gives a completely different hash.
Digital signature	Document hash encrypted with a private key; proves the signer and that the document is unchanged.
Private / public key	Secret key to sign / shared key to verify (asymmetric cryptography).
AI / ML	Machines doing intelligent tasks / learning patterns from data.
RPA	Software bots automating repetitive, rule-based computer tasks.
UPI	NPCI's 2016 real-time, 24×7, bank-to-bank payment system using a VPA / QR.
India Stack	Open DPI APIs: Presence-less, Paperless, Cashless, Consent.

Regulatory sandbox	Live testing of new FinTech products at limited scale under regulator watch (RBI 2019).
MVP	Simplest product version to test demand: Build → Measure → Learn.
P2P lending	Online platform matching borrowers with individual lenders; earns fees, lenders bear risk.
Crowdfunding	Raising small amounts from many people online: Donation, Reward, Equity, Debt.
DFI	Institutions giving long-term development finance: NABARD, SIDBI, EXIM, NaBFID.
Regulation	Authority-made rules protecting consumers, stability and fairness; enforced by licences, audits and penalties.
Why FinTechs focus on digital lending	Highest margins + huge credit gap + cheap data and rails from India Stack.
Why digital payments are popular	Free UPI rails, a billion phones, daily frequency, gateway to cross-selling.

Pair this with the study guide: it gives you the depth (frameworks, cases, numbers) for Units 1–7. This booklet tells you **how the questions are actually asked** and fills the India-specific gap.